

Branch Predictor — RTL to GDSII Flow Guide (Cadence Toolchain)

A step-by-step reference for taking the Advanced Multi-Cycle Branch Predictor from RTL through functional simulation, synthesis, and physical design using the Cadence toolchain.

Note: File paths below (e.g. PDK library locations, tool install paths) are placeholders. Replace <PDK_ROOT>, <LIB_PATH>, and similar tokens with your own environment's actual paths — these will differ depending on your lab/organization's tool installation.

Step 0: Terminal Setup & Directory Creation

```
cd ~/Desktop
mkdir branch_predictor
cd branch_predictor
```

Source your Cadence environment setup file (path depends on your local installation):

```
csh
source <CADENCE_ENV_SETUP_SCRIPT>
```

Step 1: RTL Code & Testbench Setup

Create two clean, separate .v files in your branch_predictor directory:

- Advanced_MultiCycle_Top.v — synthesizable RTL (see rtl/ in this repository)
 - tb_Advanced_MultiCycle_Top.v — testbench (see testbench/ in this repository)
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Step 2: Functional Simulation (Cadence NC-Launch)

1. Start NC-Launch:

```
nclaunch -new &
```

2. In the pop-up window:

- Click **Create cds.lib file** → Save
- Select **Don't include any libraries** → Click OK

3. **Compilation:**

- Select Advanced_MultiCycle_Top.v and click the Verilog Compiler (VLOG) icon
- Select tb_Advanced_MultiCycle_Top.v and click the VLOG icon

4. **Elaboration:**

- Expand worklib in the top-left pane
- Click tb_Advanced_MultiCycle_Top and click the Elaborator icon

5. Simulation Waveforms:

- Expand **Snapshots**
 - Right-click worklib.tb_Advanced_MultiCycle_Top and click the SimVision/Indago icon
 - Select tb_Advanced_MultiCycle_Top, click the Waveforms icon, press Play, and view the signals
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Step 3: SDC Constraints Setup

```
gedit predictor.sdc &

create_clock -name clk -period 10 -waveform {0 5} [get_ports "clk"]
set_clock_transition -rise 0.1 [get_clocks "clk"]
set_clock_transition -fall 0.1 [get_clocks "clk"]
set_clock_uncertainty 0.01 [get_ports "clk"]
set_input_transition 0.12 [all_inputs]
set_input_delay -max 0.8 [get_ports "reset"] -clock [get_clocks
"clk"]
set_output_delay -max 0.8 [all_outputs] -clock [get_clocks "clk"]
set_load 0.15 [all_outputs]
set_max_fanout 20.00 [current_design]
```

Step 4: Logic Synthesis (Cadence Genus)

Launch Genus:

```
genus
```

Run the following Tcl commands in sequence:

```
# 1. Read Technology Liberty Library
read_libs <PDK_ROOT>/lib/slow.lib

# 2. Read RTL file ONLY
read_hdl Advanced_MultiCycle_Top.v

# 3. Elaborate Top Module
elaborate Advanced_MultiCycle_Top

# 4. Read SDC Constraints (Executed AFTER elaboration)
read_sdc predictor.sdc

# 5. Set Synthesis Optimization Efforts
set_db syn_generic_effort medium
set_db syn_map_effort medium
set_db syn_opt_effort medium

# 6. Execute Synthesis Phases
syn_generic
syn_map
syn_opt

# 7. Generate Synthesis Reports
report_timing > timing_report.txt
report_power > power_report.txt
report_area > area_report.txt

# 8. Export Synthesized Netlist & Updated SDC for Innovus
write_hdl > predictor_netlist.v
write_sdc > predictor_sdc.sdc

# 9. View Schematic Diagram (Optional)
gui_show
```

Step 5: Physical Design & Layout (Cadence Innovus)

Launch Innovus:

```
innovus
```

Design Import (File → Import Design)

- **Netlist:** select predictor_netlist.v
- **LEF Files:** load technology LEF first, then macro LEF(s), from <PDK_ROOT>/lef/
 - ⚠ Order matters — the technology LEF (which defines metal layers) **must** be loaded before any macro LEF that references those layers.
- **Power Nets:** VDD | **Ground Nets:** VSS
- **Analysis Configuration:**
 - Library Sets: create slow (load slow.lib) and fast (load fast.lib)
 - RC Corners: create default_rc
 - Delay Corners: create max_delay_corner (slow + default_rc) and min_delay_corner (fast + default_rc)
 - Constraint Modes: load predictor_sdc.sdc
 - Analysis Views: setup wc (worst case → max_delay_corner, Setup analysis) and bc (best case → min_delay_corner, Hold analysis)
- Click **Save** to save globals and view files, then click **OK**

Floorplanning (Floorplan → Specify Floorplan)

- Aspect Ratio: 1.0
- Core Utilization: 0.7
- Core Margins (Left/Right/Top/Bottom): 8

Power Planning (Power → Power Planning)

1. **Global Net Connect:** connect VDD and VSS pins to power nets
2. **Add Rings:** Metal 9 (Top/Bottom) and Metal 8 (Left/Right), Width 0.5, Spacing 0.5
3. **Add Stripes:** Metal 8 (Vertical) and Metal 9 (Horizontal)
4. **Special Route:** run sroute to connect block/core pins to power rings

Physical Cells & Placement

- **Add End Caps & Well Taps:** place FILL2 pre/post cap cells; set well tap distance to 30
- **Place Standard Cells:** Place → Place Standard Cell
- **Place I/O Pins:** run place IO pins

Clock Tree Synthesis (CTS) & Timing

- Run Pre-CTS Timing Report: Timing → Report Timing (Pre-CTS)
- Execute Clock Tree Synthesis: ccopt_design (Clock menu / command line)
- Run Post-CTS Timing Report (check Setup/Hold)

Routing & Physical Checks

- Route → NanoRoute → enable Timing Driven and SI Driven → OK
- Run Post-Route Timing Analysis
- **Important:** before post-route optimization, set OCV-aware analysis:

setAnalysisMode -analysisType onChipVariation -cppr both

- If timing violations occur, optimize via ECO → Optimize Design (or optDesign -postRoute)

Export Final Netlist & GDSII

- Save Gate-Level Netlist: File → Save Netlist → save as predictor_innovus.v
 - Export GDSII: File → Save → GDS/OASIS → name output file Advanced_MultiCycle_Top.gds → OK
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Key Fixes Applied to the Original RTL

Two constructs in the original RTL required correction for clean synthesis in Genus (see inline comments in rtl/Advanced_MultiCycle_Top.v):

1. **multi_cycle_pipeline reset logic** — combining if (reset || flush) in an async-reset-sensitive block caused a CDFG-364 elaboration error in Genus. Fixed by splitting into if (reset) ... else if (flush)
2. **pattern1 initializer** — an inline register initial value (reg [10:0] pattern1 = ...) was silently ignored by Genus (VLOGPT-37), meaning the register would power up undefined in real silicon. Fixed by converting pattern1 to a localparam.